

Hayden Choi

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Education

McMaster University

Honours Computer Science Co-op, Bachelor of Applied Science (GPA: 3.8/4.0)

Expected Dec 2027

Hamilton, ON

Experience

Wealthsimple

Software Engineer Intern

Sept 2026 – Present

Toronto, ON

- Building fraud and risk features protecting **\$155B+** in assets across **4M+** clients

Trend Micro

Software Engineer Intern

May 2026 – Aug 2026

Ottawa, ON

- Built a behavioral ranking framework scoring **1,500+** risk event types on real fleet remediate-vs-dismiss outcomes, mining **8M+** human action decisions from **2,000+** enterprise tenants on a **Trino/Iceberg** lakehouse
- Surfaced an underperforming risk event on the ranking leaderboard **26 days** before customers escalated it into a **P1** product concern ticket over false alarms, ahead of any manual review or field triage
- Designed a cross-repo **CI/CD** pipeline cutting the **5 day** risk event definition onboarding down to **2 days**
- Shipped a credential-free **EKS** service across **12** AWS regions on **Pod Identity**, owning its **Terraform** footprint
- Audited a pre-release **gRPC** connector SDK, uncovering **7+** packaging, versioning, and API-contract defects

Knockri

Software Engineer Intern

Dec 2025 – Apr 2026

Toronto, ON

- Architected an end-to-end **ReAct AI** agent on IBM Watsonx Orchestrate that autonomously resolves customer support tickets, reducing ticket inbox volume by **83%** and cutting external customer support costs by **61%**
- Reduced transcription tool latency **74%** by resolving **N+1 GraphQL** queries and refactoring data-fetching architecture
- Engineered **Knockri CLEAR**, a proctoring suite validating **10k+** sessions, improving assessment integrity by **80%**
- Implemented a **Heartbeat API** via **GraphQL/UUID4** to restrict **7,000+** duplicate launches via backend locking

Brighter Signals BV

Software Engineer Intern

May 2025 – Aug 2025

Markham, ON

- Built a real-time **F1 plank wear detection** pipeline, applying baseline calibration at ingestion for **64%** better accuracy
- Developed **5+** specialized GUIs for investor demos that secured **\$1.6M+** in funding for **MVP** development
- Co-authored an **IROS 2025** paper and named a **Best Poster Finalist** for research on data-rich tactile sensors

Mana Immigration

Software Engineer Intern

May 2024 – Sept 2024

Markham, ON

- Architected **6+** full-stack Flutter apps for international clients using a Factory Pattern for reusable UI components
- Boosted app responsiveness by **50%** by implementing **Deferred Loading** for complex navigation modules

Projects

Proteus | 2nd Overall & Best Use of DigitalOcean, cuHacking7

FastAPI, Next.js, MongoDB Atlas Vector Search, Redis, DigitalOcean

Live | Devpost | GitHub

- Created a genomic IDE turning plain-English goals into scored DNA with predicted folds, accelerating wet lab testing
- Built the literature **RAG** layer, embedding **PubMed** abstracts for semantic search gated to model-edited regions

ColdApproach AI

Next.js, FastAPI, AWS (SES, SQS, Lambda), OpenAI, Hunter.io

Live | GitHub

- Built an AI outreach platform with **FastAPI** and **Next.js** using **Hunter.io** discovery and a two-pass **GPT** pipeline
- Engineered an async batch pipeline with **AWS SQS** to process up to **20** URLs through scraping and generation stages

SyllaScan

React, Python, OpenAI, AWS, FastAPI, Google Calendar API

GitHub

- Developed an **NLP pipeline** using **FastAPI** to parse unstructured PDF syllabi into validated **JSON** schemas
- Integrated **Google OAuth 2.0** and **Calendar API** to automate bulk event synchronization for academic schedules

Technical Skills

Languages: Python, Ruby, JavaScript, Java, TypeScript, C++, C, SQL

Libraries/Frameworks: Django, React, PyTorch, Scikit-learn, FastAPI, PySpark

Databases/Cloud: AWS (S3, Glue, Athena, EMR, SES, SQS), Azure, Apache Iceberg, Trino, MongoDB

Tools/Methodologies: Git, Docker, Kubernetes, Terraform, Linux, GraphQL, CI/CD, Agile